

Full Regression Test Report - KIN

Project Information

- Project: KIN Utility System
- Repository: https://github.com/anaclaireellen/IT342_G5_Naranjo_Lab1-
- Branch: vertical-slice-regression
- Date: May 10, 2026
- Platforms covered: Spring Boot backend, React web frontend, Android mobile application

Refactoring Summary

The project was refactored from technical-layer organization into vertical feature slices while preserving existing behavior. Backend authentication classes now live together in the features.auth package. Web screens and supporting utilities are grouped by feature: auth, dashboard, borrowing, messaging, profile, and shared infrastructure. Android activities were moved into auth, dashboard, and admin feature packages, while Retrofit API classes were moved into a core network package.

Regression fixes applied during the refactor:

- Updated Android registration endpoint from /api/register to /api/auth/register.
- Updated Android registration payload to send firstName and lastName, matching backend registration validation.
- Updated Android manifest activity names after package refactoring.
- Aligned AndroidX Core KTX to 1.12.0 so the mobile build remains compatible with compile SDK 34 and Android Gradle Plugin 8.3.2.
- Added clearStoredSession() removal of isAuthenticated so logout fully clears authentication state.
- Added H2 test database configuration for backend regression tests.

Updated Project Structure

```
backend/src/main/java/com/example/kin
  config/
  features/auth/
    AuthController.java
    User.java
    UserRepository.java

web/src
  features/auth/
  features/borrowing/
  features/dashboard/
  features/messaging/
  features/profile/
  shared/components/
  shared/lib/
  shared/theme.js

mobile/app/src/main/java/com/example/kin
  core/network/
```

features/admin/
features/auth/
features/dashboard/

Functional Requirements Coverage

Requirement	Backend	Web	Mobile	Automated Coverage
Register student/admin accounts	Yes	Yes	Yes	Backend auth test, mobile build
Generate CIT username/email from names	Yes	Yes	Partial preview on mobile via full-name split	Backend auth test
Login with generated email/username	Yes	Yes	Yes	Backend auth test
Persist and clear authenticated session	N/A	Yes	Yes	Web profile helper test
View dashboard by role	N/A	Yes	Yes	Web build, mobile build
Post BorrowHub requests	External Supabase	Yes	N/A	Web build
Browse and respond to requests	External Supabase	Yes	N/A	Web build
Direct messages and deal confirmations	External Supabase	Yes	N/A	Web deal-message tests
Profile settings and avatar persistence	Yes	Yes	N/A	Web profile helper test
Admin panel access	N/A	Yes	Yes	Web build, mobile build

Test Plan

ID	Feature	Test Steps / Script	Expected Result	Automation
TC-001	Backend context	Run mvn test in backend	Spring context loads with H2 test DB	KinApplicationTests
TC-002	Registration	POST /api/auth/register with first name, last name, password, role	User is saved with generated unique username and @cit.edu email	AuthControllerTest.registerGeneratesUniqueUsernameAndCitEmail
TC-003	Login	POST /api/auth/login with generated email and password	Login response returns username, role, email, and profile fields	AuthControllerTest.loginAcceptsGeneratedEmailIdentifier
TC-004	Profile lookup	GET /api/auth/profiles?usernames=...	Profile summary list is returned	AuthControllerTest.profilesReturnsRequestedProfileSummaries
TC-005	Web routing	Run npm test -- --watchAll=false --runInBand	Landing route renders after feature-slice imports	App.test.js
TC-006	Web profile session	Persist and clear profile in localStorage	Profile cache is readable and auth state is cleared	profileHelpers.test.js
TC-007	Deal messages	Create, parse, preview, and key deal messages	Deal metadata remains stable	dealMessages.test.js

ID	Feature	Test Steps / Script	Expected Result	Automation
TC-008	Web production build	Run npm run build in web	React app compiles successfully	Build command
TC-009	Mobile unit/build regression	Run gradlew testDebugUnitTest assembleDebug with Android SDK env vars	Unit tests pass and debug APK assembles	Gradle command

Automated Test Evidence

Command	Result
mvn test	Passed: 4 tests, 0 failures, 0 errors
npm test -- --watchAll=false --runInBand	Passed: 3 suites, 5 tests
npm run build	Passed: production build compiled successfully
gradlew testDebugUnitTest assembleDebug	Passed: 38 Gradle tasks, debug APK assembled

Generated evidence locations:

- Backend Surefire reports: backend/target/surefire-reports/
- Web production build: web/build/
- Mobile unit test report: mobile/app/build/reports/tests/testDebugUnitTest/
- Mobile debug APK: mobile/app/build/outputs/apk/debug/

Regression Test Results

Area	Status	Notes
Backend API	Passed	Auth controller and Spring context tests pass.
Web frontend	Passed	Jest tests and production build pass after vertical-slice import updates.
Mobile app	Passed	Unit tests and debug assembly pass after SDK env setup and AndroidX version alignment.
External services	Not executed end-to-end	Supabase-backed BorrowHub and messaging require live Supabase data and credentials; covered by build and utility tests only.

Issues Found

Issue	Impact	Fix Applied
Android client posted registration to /api/register	Mobile registration could not reach current backend endpoint	Changed Retrofit route to /api/auth/register
Android registration payload used username instead of required first/family names	Backend validation rejected mobile registration	Split full name into firstName and lastName payload fields

Issue	Impact	Fix Applied
Android package moves broke manifest references	App launch/navigation would fail	Updated manifest and imports to feature packages
AndroidX Core KTX 1.16.0 incompatible with AGP 8.3.2/compile SDK 34	Mobile build failed before compilation	Downgraded Core KTX to 1.12.0
Web logout helper left isAuth in storage	Local session could remain marked authenticated	Removed isAuth in clearStoredSession()
Backend tests depended on local MySQL	Regression tests were environment-sensitive	Added H2 test database configuration

Fixes Applied

All identified regressions from the refactor and test runs were fixed. Final automated regression status is passing for backend, web, and mobile build/unit coverage.